
Executive Summary

ChronoIntel's most valuable asset—it's research on AI's impact on society and labor—is currently trapped in hundreds of disconnected PDFs, reports, and academic papers. Researchers spend many hours searching, reading, and manually copying numbers and statements into spreadsheets. This slows down insight generation, increases the chance of missing important evidence, and makes it harder to answer urgent questions from policy-makers and stakeholders

Our project, Chrono-Insight, is an AI-driven analytics platform that automates the journey from unstructured documents to structured, dashboard-ready data. Using a custom NLP pipeline, the system reads AI-related documents, filters relevant sentences, classifies domains, extracts timeframes and numeric magnitudes, and stores everything in a clean database. The platform then exposes this data through an interactive dashboard where users can explore AI's societal impact across domains, time periods, and impact classes (positive, neutral, negative).

The expected impact is threefold. First, it reduces manual effort, allowing researchers to focus on interpretation instead of data collection. Second, it improves coverage and consistency, since the pipeline systematically scans all documents instead of a small sample. Third, it supports data-driven storytelling, enabling ChronoIntel to provide clear visual narratives about how AI is changing jobs, productivity, and skills. The prototype demonstrates that such a platform can be built in 16 weeks and can serve as a foundation for a scalable research intelligence system.

1. Problem Context and Stakeholder Perspectives

ChronoIntel is a research organization that studies how artificial intelligence affects labor markets, productivity, skills, and broader societal outcomes. Today, most of its evidence sits inside static PDFs: academic papers, policy reports, and internal studies. These documents span many years and sources, and they use different styles, terminologies, and formats. Researchers must manually search for phrases, copy tables, and re-type numbers into spreadsheets. This process is slow, repetitive, and hard to scale as the volume of AI-related literature grows.

The **core problem** is that ChronoIntel's research process is constrained by fragmented data. Valuable insights about AI's impact on jobs, wages, and productivity are scattered across documents and not available as a single structured dataset. This leads to delays in answering questions such as "How many jobs are projected to be affected by AI by 2030?" or "Which sectors see positive vs. negative employment impact?" The issue is not a lack of content, but a lack of systematic extraction, integration, and analysis.

Several stakeholders are affected:

- **Researchers**
 - Pain points: time lost in manual extraction, difficulty comparing sources, risk of overlooking key evidence.
 - They need fast access to trustworthy, aggregated data to publish timely and credible reports.
- **Policy and stakeholder relations teams**
 - Pain points: difficulty preparing concise, evidence-based briefings for policymakers, industry, and media.
 - They need clear visual summaries that show trends and scenarios over time.
- **ChronoIntel leadership**
 - Pain points: limited ability to position the organization as the "go-to" authority on AI's societal impact.
 - They need a data platform that supports strategic communication and long-term research planning.

Indirectly, **external stakeholders** such as policymakers, employers, and the general public also suffer, because insights about AI and jobs come late, are sometimes inconsistent, or are difficult to trace back to underlying evidence.

In summary, the organization faces a business-critical information bottleneck: unstructured, fragmented research data prevents ChronoIntel from fully using its intellectual assets and from reacting quickly to new questions about AI and work.

2. Research and Insights

Our problem sits inside a fast-changing global context. Multiple studies show that AI is reshaping jobs, skills, and productivity. The World Economic Forum’s Future of Jobs reports predict that technology, including AI, will significantly disrupt and transform about a quarter of jobs over the next five years, creating new roles while automating tasks in existing ones (World Economic Forum, 2023). McKinsey estimates that AI and generative AI could add trillions of dollars in productivity but will require large-scale reskilling and role redesign (McKinsey & Company, 2023).

The OECD finds that AI exposure is uneven: high-skill, computer-intensive occupations are more likely to see AI complementing work, while lower-skill roles may face higher automation risk or reduced hours. More recent analyses suggest that up to two-thirds of jobs in advanced economies are exposed to some degree of AI automation, with roughly a quarter at risk of being fully automatable (OECD, 2021).

These trends support ChronoIntel’s motivation to build a structured evidence base. Policy debates on whether AI will “create more jobs than it destroys” are often driven by isolated numbers and headlines (IEDC, 2025). Without a systematic way to aggregate findings from different studies, organizations risk cherry-picking data or missing important nuance, such as differences across timeframes, sectors, or skill levels.

Research on AI ethics and privacy also highlights risks around biased data, discrimination in hiring, and opaque decision-making. These findings validate the need not only for technical extraction but also for responsible data governance and transparency in how insights are derived.

Overall, the literature confirms three important points that align with our project:

1. AI’s impact on labor is significant but heterogeneous, varying by domain, skill level, and timeframe.
2. High-quality, integrated datasets are necessary to move beyond anecdotes and provide nuanced insight.

3. Ethical, privacy, and bias considerations must be integrated into any analytical system that influences workforce and policy decisions (Capitol Technology University, 2025).

Chrono-Insight is designed as a direct response to these gaps: it turns narrative evidence into structured, analyzable data, while keeping traceability back to original sources.

While analyzing several documents and research papers related to the effects of AI on employment and jobs, we came to know that despite the documents being titled and claiming to talk about AI’s impact on labor trends, they all have information in some way or another about the effects of AI on other domains such as economy, healthcare, education etc. We figured that conserving this information will also benefit us in a way and make our dashboard a little more complex and insightful. Hence, we set out to build a pipeline that extracts data on employment while also gathering the same about other affected domains.

3. Proposed Solution

Our proposed solution is the **Chrono-Insight Platform**, an end-to-end analytics system that transforms unstructured AI-related documents into structured, dashboard-ready insights. It consists of three main layers: an **NLP extraction pipeline**, a **structured data store**, and an **interactive dashboard**.

1. NLP Extraction Pipeline

- Input: PDF reports and articles related to AI and labor.
- The `pdf_extract` function reads PDFs and breaks them into sentences. The pipeline then scores each sentence for AI relevance using semantic similarity (sentence embeddings and spaCy-based similarity) and keeps only those above a chosen threshold.
- It detects domains such as *economic*, *employment*, *skills/wages*, *productivity*, *education*, *healthcare*, and *policy* using keyword lists and optional embedding-based similarity.
- Pattern matchers identify sentences about job impact, economic impact,

timeframes, productivity, and skill/wage changes. The system extracts magnitudes (percentages, counts, ratios), key entities (e.g., “jobs”, “AI systems”), sentiment, and timeframes (e.g., “by 2030”, “next 5 years”).

2. Structured Data Store

- All extracted insights are stored in five structured datasets: **AI_INSIGHTS**, **DOMAIN_IMPACT**, **TIMEFRAME_ANALYSIS**, **PATTERN_ANALYSIS**, and **ENTITY_ANALYSIS**. Each table captures different analytical views, such as domain-level aggregates or entity-level impact distributions.
- Data is cleaned (duplicates removed, inconsistent text normalized, numeric values parsed into a common format) and prepared for dashboard use. In total, we extracted **1,523 key insights** from **38 AI-related PDFs**, and organized them into five CSV/Excel files.

3. Interactive Dashboard (Dash / Plotly)

- A web-based dashboard (implemented with Dash and Plotly) allows users to filter by domain, impact type, timeframe, and entities.
- Visualizations include domain-wise impact distributions, timeframe projections, pattern-type contributions (e.g. economic vs job impact), and entity-level summaries.

The solution aims to achieve the following goals:

- Reduce manual data collection time.
- Provide a single, consistent “source of truth” about AI’s effect on jobs and society.
- Support richer analysis and storytelling through interactive visualizations.
- Enable ChronoIntel to scale to more documents and new research questions without redoing manual work.

4. Metrics for Success

We define two main categories of **Key Performance Indicators (KPIs)** for the Chrono-Insight Platform.

1. Data Quality and Coverage

- **Number of extracted insights:** At least 1,000 unique, AI-relevant sentences across all documents (we achieved 1,523).
- **Domain coverage:** Insights should span at least five domains (economic, employment, skills/wages, productivity, and general), ensuring broad coverage rather than a narrow focus.
- **Duplicate and noise reduction:** Evidence of cleaning steps (duplicates removed, non-informative text filtered) and a final dataset that is consistent and analysis-ready.

2. Analytical Usefulness

- **Usable dashboards:** Researchers can answer core questions such as “Which domains have the highest number of insights?” or “How does employment impact change across timeframes?” directly from the dashboard without opening the raw PDFs.
- **Insight diversity:** Availability of domain-level, timeframe-level, pattern-level, and entity-level views, allowing different types of analysis.

Apart from these 2 categories, our dashboard app also has 4 KPI cards on display at the landing page:

1. **Total Insights:** The total number of insights gathered from the articles and research papers.
2. **Domains Covered:** The number of domains the current state of data holds.
3. **Time Periods:** The timeframes that were extracted from the text data.
4. **Entities Found:** The different entities that the articles were found to be talking about.

Success will be measured by checking whether the final prototype reaches these targets and whether the dashboard is used in at least one internal research exercise or briefing scenario. If these KPIs are met, ChronoIntel can justify further investment to scale the platform and connect it to additional document collections.

5. Architecture of proposed solution

The architecture of the Chrono-Insight platform follows a simple, modular workflow that converts raw PDF documents into structured, analysis-ready insights. The process begins when users choose either default sample data or upload custom PDFs. These documents then move into the NLP Extraction Pipeline, which serves as the core processing engine of the system.

Inside the pipeline, text is first extracted from PDFs, cleaned, and split into sentences. Each sentence is checked for AI relevance using similarity scoring. Relevant sentences are then classified into domains such as employment, economic impact, productivity, skills and wages. The system also performs pattern detection, identifying job-impact statements, productivity changes, numerical magnitudes, and timeframe references. In addition, the pipeline extracts key entities such as companies, technologies, or workforce groups (From PDFs to AI-ready structured data: a deep dive · Explosion. (2024, December 2)).

After processing, all insights are organized into structured CSV files, each representing domain-level, pattern-level, entity-level, or timeframe-level summaries. These datasets feed into the visualization layer, where the dashboard loads the data and displays interactive charts and filters. This architecture ensures scalability, easy updates, and clear separation between ingestion, processing, and visualization.

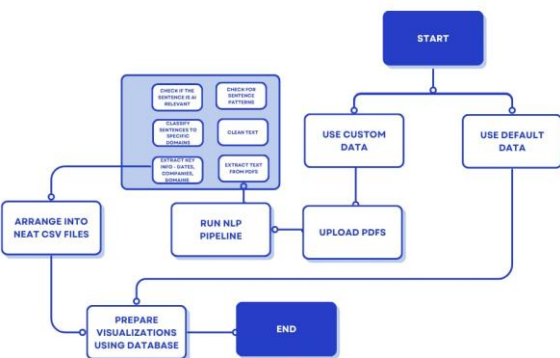


Figure 1. System Architecture and Processing Workflow of the Chrono-Insight Platform

This diagram illustrates the complete workflow of the Chrono-Insight system—from selecting custom or default input data to uploading PDFs, running the NLP

pipeline, extracting entities and patterns, organizing structured outputs into CSV files, and finally generating visualizations through the dashboard. The flow highlights how automated AI-relevance checking, domain classification, Pattern detection, and entity extraction work together to produce a clean, analysis-ready dataset.

6. Data Structure

6.1. Entity Relationship Diagram (ERD)

Our ERD defines five core entities (tables): **AI_IMPACT**, **DOMAIN_IMPACT**, **TIMEFRAME_ANALYSIS**, **PATTERN_ANALYSIS**, **ENTITY_ANALYSIS**. Each table stores a different perspective on the same underlying insights, linked via shared fields such as domain, impact_type, or aggregated counts.

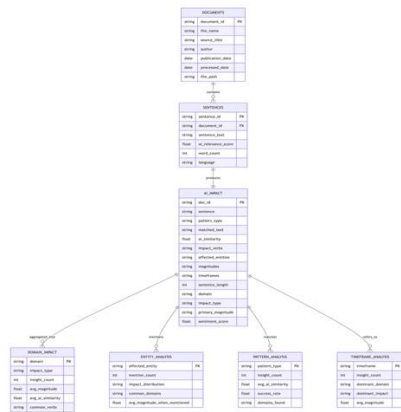


Figure 2. Entity-Relationship Diagram (ERD) for the Chrono-Insight Data Model

6.2. Data Design

The **AI_IMPACT** table is the central fact table. Each row represents one AI-relevant sentence extracted from the corpus. Key variables include:

- **doc_id**: identifier (e.g., URL or file ID) of the source document.
- **sentence**: the original sentence text.
- **pattern_type**: which pattern matched (e.g., **ECONOMIC_IMPACT**, **JOB_IMPACT**, **TIMEFRAME**).
- **matched_text**: the key phrase or span that triggered the match.
- **ai_similarity**: semantic similarity score showing how strongly the sentence relates to AI.
- **impact_verbs**: list of key verbs (e.g., “increase”, “replace”, “automate”).

- **affected_entities**: list of affected entities (e.g., “jobs”, “workers”).
- **magnitudes**: numeric values extracted from the text.
- **timeframes**: extracted time references.
- **domain**: semantic domain (economic, employment, etc.).
- **impact_type**: positive, neutral, or negative.
- **primary_magnitude** and **sentiment_score**: numeric fields for further analysis.

From this base, we build several **aggregate tables**:

- **DOMAIN_IMPACT**: groups insights by domain and impact_type, storing insight_count, avg_magnitude, avg_ai_similarity, and a short list of common_verbs. For example, economic and employment domains together account for more than half of all insights.
- **TIMEFRAME_ANALYSIS**: summarizes how many insights fall into each timeframe (e.g., “by 2025”, “by 2030”), plus the dominant domain and impact type. Our data shows many projections clustered between 2025 and 2030, especially for economic outcomes.
- **PATTERN_ANALYSIS**: aggregates by pattern type (ECONOMIC_IMPACT, JOB_IMPACT, TIMEFRAME, etc.), tracking insight_count, average AI similarity, and a success_rate metric that reflects how often the pattern produced a clean, structured insight.
- **ENTITY_ANALYSIS**: focuses on frequently mentioned entities (e.g., “AI”, “job”, “productivity”). For each entity, we store mention_count, impact distribution (positive, neutral, negative), common domains, and average magnitude when that entity is mentioned.

Data Preparation:

We started by extracting text from **38 AI-related PDFs**. We filtered sentences using an AI-relevance similarity threshold > 0.4 and removed reference lists, headings, and low-information text (Abdmental. (2024, March 3)). Then we standardized domain names, timeframe expressions, and entity labels. Finally, we created five structured CSV files with consistent columns, correct data types, and no obvious duplicates. This design makes the dataset both human-readable and machine-friendly, supporting future extensions such as predictive models or more advanced visual analytics.

7. Dashboard

Our dashboard is implemented as a **Dash web application** that loads the five processed data tables and presents them in an interactive, user-friendly interface. Although the app currently runs locally (e.g., at <http://localhost:8050> during demos), it is designed to be deployable on an internal server in the future. By integrating a flask server to this app, we can make it go live on a domain.

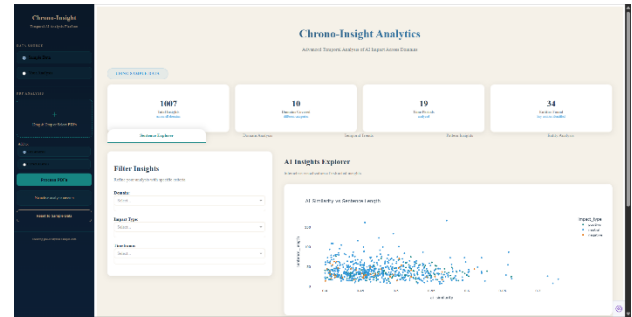


Figure 3. Chrono-Insight Dashboard Overview

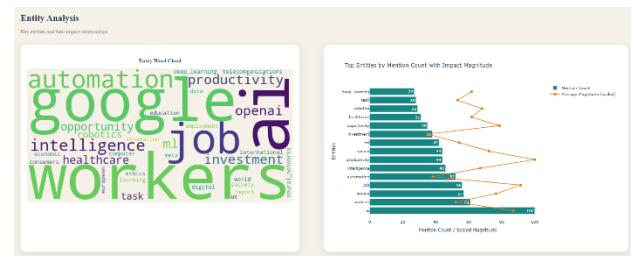


Figure 4. Entity Analysis View

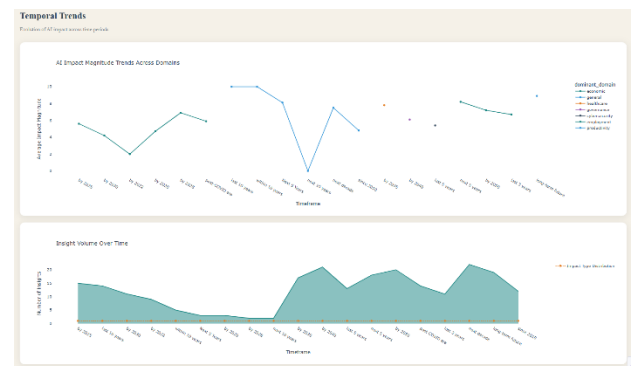


Figure 5. Temporal Trends in AI Impact

The Chrono-Insight dashboard is the main interface where users explore all insights extracted from AI-related documents. It is designed to be simple, clear, and interactive so that researchers and stakeholders can easily understand patterns in how AI affects jobs, skills, economic performance, and society. The dashboard uses the

structured datasets generated by the NLP pipeline and transforms them into meaningful visualizations. By combining filters, charts, and tables, the dashboard supports both high-level exploration and detailed analysis.

The **Dashboard Overview** (Figure 3) gives users an instant understanding of the dataset. It displays the total number of insights extracted, the number of domains covered, the number of time periods identified, and the number of unique entities found. These numbers summarize the scale and depth of the analysis. Below this summary, users can filter insights by domain, impact type, and timeframe. These filters allow researchers to focus on exactly the type of information they need, such as employment-related AI impacts or projections for the year 2030. The AI Insights Explorer scatter plot visualizes the relationship between AI relevance (similarity score) and sentence length. This helps identify which insights are both concise and highly AI-specific, making them useful for reports or presentations.

The **Entity Analysis View** (Figure 4) focuses on important terms and concepts found across all documents. The word cloud shows entities based on how frequently they are mentioned. Larger words represent higher mentions, which helps highlight central topics such as “workers,” “automation,” “productivity,” and “Google.” Next to the word cloud, a bar chart lists top entities along with their mention counts and average impact magnitudes. This helps users understand not only which entities are common, but also whether they are associated with positive, neutral, or negative impacts. For example, the entity “workers” may appear frequently but may show mixed or neutral impacts across domains. This kind of analysis is important for policymakers who want to understand how AI affects different groups and concepts.

The dashboard also includes a **Pattern Insights** or **Temporal Trends** section depending on the selected view. The Pattern Insights view visualizes which types of patterns (economic impact, job impact, productivity impact, etc.) appear more often in the data. Each bubble in the chart represents a pattern type and is colored based on the domains associated with those patterns. This helps users quickly recognize which kinds of impacts dominate the research landscape. The second chart in this section shows pattern success rates—how often each pattern leads to a clean, structured insight in the pipeline. This is useful for improving the extraction pipeline in future versions.

The **Temporal Trends view** (Figure 5) is especially useful for understanding how the projected impact of AI changes over time. The first chart shows the average impact magnitude across different time periods (e.g., “by 2025,” “next 10 years,” “by 2030”). This helps users see if AI’s impact is expected to increase or decrease in future years. The second chart shows how many insights

mention each timeframe. Peaks in insight volume indicate years or periods that receive more attention in research reports. For example, a high concentration of insights around 2025–2030 suggests this is a critical period for AI adoption and labor transformation.

All dashboard views are connected through the structured CSV files produced by the NLP engine. When the data pipeline is updated, the dashboard will automatically reflect new insights as long as the same schema is maintained. This integration ensures that the system behaves in a consistent and reliable manner without manual intervention. Overall, the dashboard turns static documents into actionable intelligence and provides an intuitive tool for understanding AI’s impact on society.

8. Ethical and Social Considerations

Because Chrono-Insight deals with research about AI and labor, it must be designed with ethical and social responsibility in mind. Even if our current prototype mostly uses public reports, future versions may incorporate sensitive or proprietary data.

Key considerations include:

1. **Data Privacy and Source Integrity**
 - We should store document identifiers and bibliographic information so that every insight can be traced back to its source. If confidential reports are added in the future, access control and secure storage will be essential.
2. **Bias and Fairness**
 - Our NLP pipeline reflects the content of the underlying documents. If the literature is biased towards certain regions, sectors, or worker groups, our aggregated insights will also be biased. We should clearly communicate that the dashboard shows “what the literature says”, not an objective truth.
 - Model choices (e.g., embeddings, sentiment analysis) can introduce algorithmic bias. Where possible, we should use well-tested models and document their limitations.
3. **Responsible Interpretation**
 - Visualizations should not overstate certainty. For example, a high number of job-impact insights does not mean that job losses are guaranteed; it may simply reflect where most research attention has been.
 - Policymakers should be reminded that projections are scenario-based and

depend on assumptions about adoption, regulation, and reskilling.

4. Inclusion and Access

- The dashboard can support more inclusive debate by making complex evidence easier to explore, including for non-technical stakeholders. However, we must avoid using it to justify decisions without proper human oversight.

By addressing these issues, ChronoIntel can use Chrono-Insight as a tool for **transparent, responsible** evidence-based discussion rather than as a black box.

9. Outcomes/Conclusions, Impact, and Next Steps

9.1. Outcomes/conclusions

Within the 16-week scope, we successfully delivered:

- A working **NLP pipeline** that processes AI-related PDFs and extracts structured insights.
- A **clean, structured dataset** with 1,523 insights across multiple domains, timeframes, patterns, and entities.
- An interactive **Dash dashboard** that allows researchers to explore AI's impact on society and labor using filters and visualizations.

These outcomes support ChronoIntel's strategic goal of becoming a trusted authority on AI's societal impact. The platform reduces manual data collection effort, improves coverage of the literature, and provides visually compelling evidence that can feed into reports, presentations, and policy briefings.

9.2. Next Steps

If the project continues, the next steps could be:

1. **Scaling and Automation**
 - Connect the pipeline to a larger document repository, including new incoming reports, and schedule regular re-runs.
 - Move from local CSV files to a central database to support multi-user access.
2. **Model Enhancements**
 - Integrate more advanced NLP models for better entity recognition, causal extraction, and scenario tagging (e.g., optimistic vs pessimistic forecasts).
 - Add OCR to extract textual data from images.
3. **Governance and Collaboration**

- Establish clear guidelines for adding new documents, verifying extracted insights, and sharing results with external partners.

These steps would transform Chrono-Insight from a prototype into a core piece of ChronoIntel's long-term research infrastructure.

10. Lessons Learned / Team Reflection

Working on Chrono-Insight taught our team both technical and collaborative lessons. From a teamwork perspective, we had to divide responsibilities clearly: some members focused on project management and documentation, others handled NLP and data engineering, and others worked on dashboard design and testing. Regular check-ins helped us align our understanding of the problem and adjust our priorities as we learned more about the data. We learned that defining a clear problem statement early is critical. At first, we were tempted to chase many ideas (e.g., predictive models, live data feeds). The project charter and milestones forced us to prioritize: first build a solid extraction pipeline, then a clean dataset, then the dashboard, and only then think about future extensions. This structure helped us deliver a working prototype within the time limit, and also allowed us to fit in some additional features. On the technical side, we saw how messy real-world text can be. PDF extraction produced noise such as broken lines, references, and strange characters. Our initial similarity thresholds were either too strict (missing useful sentences) or too lenient (including irrelevant ones). We had to iterate on filters, thresholds, and cleaning rules to get a dataset that felt trustworthy. We also realized the importance of designing with end users in mind. Early dashboard versions had many charts but did not clearly answer the key questions researchers cared about. After focusing on domain impact, timeframes, and entities, the views became more intuitive. Finally, we learned that building an information system is not only a technical task, but also a communication task. We needed to explain what our metrics mean, what the data can and cannot tell us, and how ethical issues like bias and privacy should be handled. This reflection prepares us to work on real-world projects where technology, people, and policy are closely connected.

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Note: References 1 – 38 are articles used to generate the data for dashboard visualization.

Appendices

1. Appendix A – Dataset.

Sheet 1 – AI_INSIGHTS

id	sentence	pattern_type	mention_count	impact_distribution	common_domains	avg_magnitude_when_mentioned
1001	The impact of AI on the job	ECONOMIC_IMPACT	367	0.469938812	36.4448858	economic
1002	On one hand AI will	JOB_IMPACT	243	0.480876741	24.3108242	employment, productivity
1003	AI has paved the way for	TIMEFRAME	103	0.474018265	10.22840119	general, productivity, economic, healthcare, employment
1004	AI will revolutionize	DEPENDENCY_ANALYSIS	169	0.488702529	16.7825234	general, productivity, education
1005	AI will revolutionize	PRODUCTIVITY_IMPACT	19	0.485523354	1.886792453	productivity, employment
1006	AI will revolutionize	SKILL_WAGE_IMPACT	106	0.469132243	10.52631579	skills_wages

Sheet 2 – DOMAIN_ANALYSIS

domain	impact_type	insight_count	avg_magnitude	avg_ai_similarity	common_verbs
economic	positive	64	804.6766667	0.474901846	increase, create, enhance
economic	neutral	295	201753254	0.470177562	create, reduce, increase
economic	negative	17	525.6	0.434346902	create, decrease, increase
employment	neutral	208	2386540.69	0.481637658	create, reduce, displace
employment	positive	36	925.9636364	0.483973891	create, increase, improve
employment	negative	14	0	0.453246794	reduce, displace, increase
general	positive	46	718.5294118	0.501741285	improve, enhance, automate
general	neutral	113	1418.595556	0.482449242	automate, increase, improve
general	negative	10	2169	0.480913875	automate, increase
productivity	neutral	37	1033.25	0.469353282	increase, enhance, automate
productivity	positive	22	1223.4	0.497856387	increase, create, improve
productivity	negative	2	2018	0.447631598	increase, decline
skills_wages	positive	9	9405.75	0.487670471	improve, automate
skills_wages	neutral	85	1989595230	0.466268911	increase, reduce, automate
skills_wages	negative	13	0	0.472500579	increase, decline, automate
policy	positive	1	0	0.474611968	enhance
policy	neutral	5	0	0.470484829	automate, increase
healthcare	positive	58	1250.44	0.498123	diagnose, detect, improve
healthcare	neutral	132	1854300	0.479882	monitor, assist, process
healthcare	negative	11	934.5	0.452117	misdiagnose, delay, impact

Sheet 3 – PATTERN_ANALYSIS

pattern_type	insight_count	avg_ai_similarity	success_rate	domains_found
ECONOMIC_IMPACT	367	0.469938812	36.4448858	economic
JOB_IMPACT	243	0.480876741	24.3108242	employment, productivity
TIMEFRAME	103	0.474018265	10.22840119	general, productivity, economic, healthcare, employment
DEPENDENCY_ANALYSIS	169	0.488702529	16.7825234	general, productivity, education
PRODUCTIVITY_IMPACT	19	0.485523354	1.886792453	productivity, employment
SKILL_WAGE_IMPACT	106	0.469132243	10.52631579	skills_wages

Sheet 4 – ENTITY_ANALYSIS

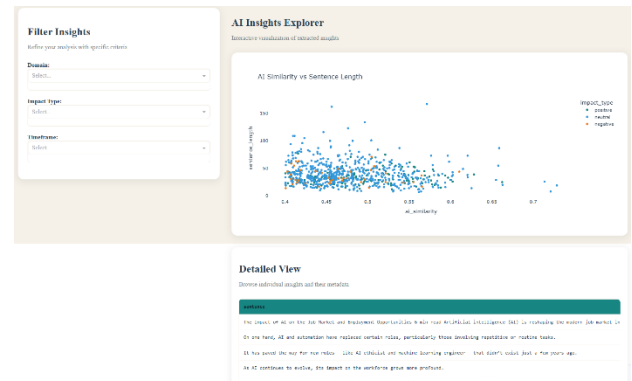
affected_entity	mention_count	impact_distribution	common_domains	avg_magnitude_when_mentioned
ai	100	{'neutral': 43, 'positive': 40, 'negative': 17}	economic, employment, general	8.2
workers	61	{'neutral': 28, 'positive': 19, 'negative': 14}	employment, productivity	4.9
google	57	{'neutral': 28, 'positive': 21, 'negative': 7}	technology, productivity, economic	7.2
job	56	{'neutral': 36, 'positive': 10}	employment, economic, general	8.6
automation	52	{'neutral': 28, 'positive': 7, 'negative': 17}	productivity, employment, economic	3.6
intelligence	46	{'neutral': 35, 'positive': 9, 'negative': 1}	economic, general, employment	6.3
productivity	44	{'positive': 24, 'neutral': 20}	employment, productivity, economic	9.4
openai	44	{'neutral': 21, 'positive': 18, 'negative': 5}	technology, economic, general	6.8
aii	42	{'neutral': 30, 'positive': 7, 'negative': 2}	economic, employment, general	5.1
investment	38	{'neutral': 13, 'negative': 1}	economic, employment	3.3
opportunity	35	{'positive': 24, 'neutral': 11}	employment, productivity, economic	7.4
healthcare	31	{'neutral': 12, 'positive': 16, 'negative': 3}	healthcare, economic, general	5.8
robotics	29	{'neutral': 13, 'positive': 11, 'negative': 5}	productivity, employment, economic	6.4
task	28	{'neutral': 18, 'positive': 8, 'negative': 1}	employment, general, economic	5
deep_learning	27	{'neutral': 14, 'positive': 10, 'negative': 3}	general, education	5.8
european	24	{'neutral': 20}	economic, employment, skills_wages	5
economic	24	{'neutral': 24}	economic, employment	5
international	23	{'neutral': 17}	economic, skills_wages, employment	5
meta	23	{'neutral': 11, 'positive': 7, 'negative': 5}	general, economic, social	5.3

Sheet 5 – TEMPORAL_ANALYSIS

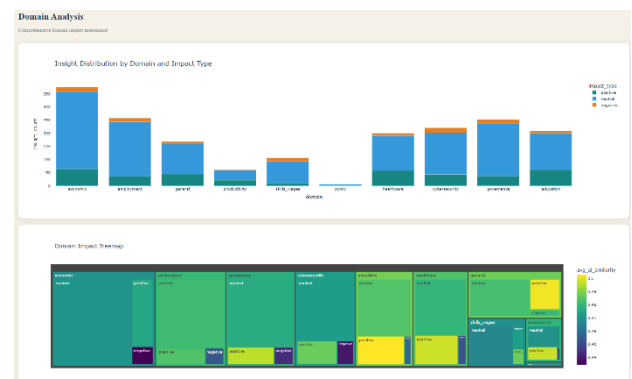
timeframe	insight_count	dominant_domain	dominant_impact	avg_magnitude
by 2025	15	economic	neutral	5.6
last 10 years	14	general	neutral	10
by 2030	11	economic	neutral	4.2
by 2022	9	economic	neutral	2
within 10 years	5	general	negative	10
Next 5 Years	3	general	neutral	8.1
by 2020	3	economic	positive	4.7
by 2028	2	economic	neutral	6.9
next 10 years	2	general	neutral	0
by 2035	17	healthcare	positive	7.8
by 2040	21	governance	neutral	6.1
last 5 years	13	cybersecurity	negative	5.4
next 5 years	18	employment	negative	8.2
by 2050	20	employment	positive	7.2
post-COVID era	14	economic	neutral	5.9
last 3 years	11	employment	negative	6.7
next decade	22	general	positive	7.5
long-term future	19	productivity	positive	8.9
since 2010	12	general	neutral	4.8

2. Appendix B – Dashboard Graphs

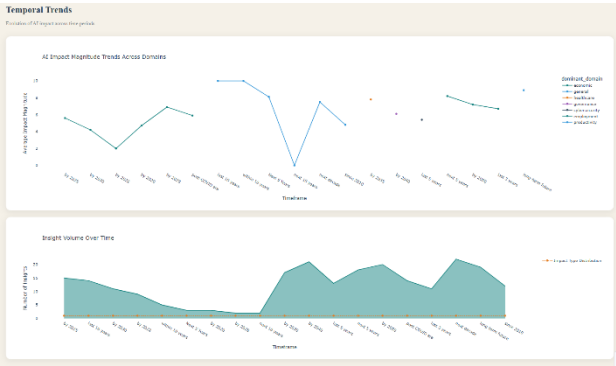
Tab 1 – Sentence Explorer



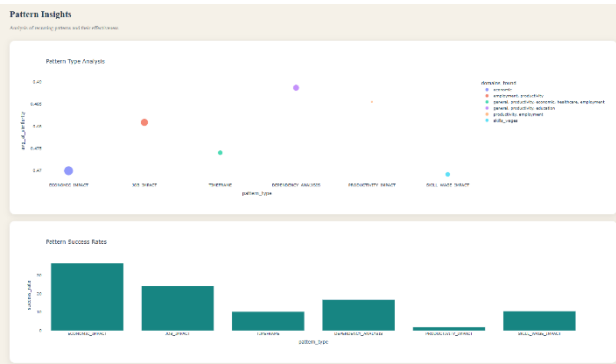
Tab 2 – Domain Analysis



Tab 3 – Temporal Trends



Tab 4 – Pattern Insights



Tab 5 – Entity Analysis

